Matrix inequalities and norms

Corrected generalized geometric-mean product conjecture

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: The four interacting exponents make this corrected conjecture challenging; its immediate consequences concern specialists in matrix means and log-majorization.

Resolution — 2026-09-11

Negative result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS.**

Rational positive definite 3×3 matrices with $r = s = 1$, $p = 2$ and $t = 1/8$ violate the corrected eigenvalue log-majorization conjecture. An exact integer-power construction and rational norm separation establish failure of the first ordered eigenvalue inequality.

The exact target is resolved. The original statement and source evidence are retained below; its former difficulty rating is historical.

Primary manuscript: complete proof PDF, standalone TeX, Theorem 1.1 and its proof; authorship and scope. The independent review checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. Submission record.

Problem statement

For positive definite $A, B \in \mathbb{C}^{n\times n}$, real r, and $t \in [0,1]$, define

$$G_{r,t}(A,B) = A^{r/2}(A^{-1/2}BA^{-1/2})^t A^{r/2}.$$

Is it true, for every $n \geq 1$, every such A, B, every $p \geq 1$, $t \in [0,1]$, and either $r, s \geq 1$ or $r, s \leq 0$, that

$$\lambda(G_{r,t}(A,B)^p G_{s,1-t}(A,B)^p) \prec_{\log} \lambda(A^{p(r+s-1)}B^p)?$$

The eigenvalues of these products are positive real numbers, ordered decreasingly: each product is similar to a positive definite matrix. For positive decreasing vectors $x, y \in \mathbb{R}^n$, $x \prec_{\log} y$ means $\prod_{j=1}^k x_j \leq \prod_{j=1}^k y_j$ for $1 \leq k < n$, with equality at $k = n$.

This compares nonlinear matrix-mean products with products of powers of the input matrices. It would give simultaneous multiplicative control for every partial product of ordered eigenvalues and strengthen norm estimates used with positive definite matrix functions.

References

1. M. M. Ghabries, H. Abbas, B. Mourad and A. Assi, *New log-majorization results concerning eigenvalues and singular values and a complement of a norm inequality*, preprint (2021), Conjecture 2.1, p. 6. PDF.
2. M. M. Ghabries, *Contributions to Matrix Inequalities and Some Applications*, PhD thesis, University of Angers and Lebanese University (2022), final Open Problems, Problem 6, pp. 109–110. Thesis uploaded by its author.
3. M. M. Ghabries, *A log-majorization inequality for normal matrices with applications to determinantal inequalities and geometric means* (2026), §4. Full text.

Status check (2026-09-10): Theorem 2.1 and equation (13) of the first source prove a substantive included region: $r, s \geq 1$, $1 \leq p \leq 2$, and $\frac{r(p-1)}{(r+s)p} \leq t \leq \frac{rp+s}{(r+s)p}$. This includes $p = 1$ for all t. The full parameter range is the remaining question. The first source refutes its earlier unrestricted singular-value Conjecture 1.2; the statement here is the explicitly proposed replacement. The thesis retains it. The July 2026 paper concerns eigenvalue products of ordinary weighted means and does not claim this full r, s, p statement. Current arXiv versions and searches for the authors, "Conjecture 2.1 generalized geometric mean" and 2025/2026 yielded no full resolution. This is a bounded check, not certification of openness.